

Name:

ID #

McGill University

STATISTICS (MATH324)

Mid-Term Exam

Duration: 5 hours

Available Friday Oct 16th to Monday Oct 19th, 2020

Instructions: This test consists of four questions. There is no optional question. Please answer all four questions. This is an open book exam.

- $X \sim \text{Normal}(\mu, \sigma^2) :$ $f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$ $x \in \mathbb{R}$
- $X \sim \text{Binomial}(n, p) :$ $p_X(x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \dots, n$
- $\text{Bernoulli}(p) = \text{Binomial}(1, p)$
- For $Z \sim N(0, 1)$, $P(|Z| < 1.96) = 0.95$.

Good luck.

Question 1. (*Sample size determination and destructive sampling*)

Destructive Sampling, in which the test to determine whether an item is defective destroys the item, is generally expensive, and the high costs involved often prohibit large sample sizes. For example, suppose the National Highway Safety Administration (NHSA) wishes to determine the proportion of new tires that will fail when subjected to hard braking at a speed of 60 miles per hour. NHSA can obtain the tires for \$25 (wholesale price) each. Suppose the budget for the experiment is \$10,000, and NHSA wishes to estimate the percentage that will fail to within 0.02 with 95% confidence. Assuming that the entire \$10,000 can be spent on tires (ignoring other costs) and that the true fraction that will fail is approximately 0.05, can NHSA attain its goal while staying within the budget? Why? **(10 marks)**

Solution:

Associated with each tire we have a binary random variable

$$X = \begin{cases} 1, & \text{if it fails,} \\ 0, & \text{otherwise,} \end{cases}$$

where $p = P(X = 1)$. Let $\hat{p}_n$ be the sample proportion of tires that fail. We want to choose n , the sample size, such that

$$P(|\hat{p}_n - p| \leq 0.02) = 0.95.$$

For large n ,

$$Z = \frac{\hat{p}_n - p}{\sqrt{\frac{p(1-p)}{n}}} \stackrel{app}{\sim} N(0, 1),$$

and hence

$$0.95 = P(|\hat{p}_n - p| \leq 0.02) = P\left(\frac{\hat{p}_n - p}{\sqrt{\frac{p(1-p)}{n}}} \leq \frac{0.02}{\sqrt{\frac{p(1-p)}{n}}}\right) \approx P(|Z| \leq 1.96).$$

Thus using

$$\frac{0.02}{\sqrt{\frac{p(1-p)}{n}}} = 1.96$$

and replacing p by 0.05, we find

$$n = (0.05)(1 - 0.05) \left(\frac{1.96}{0.02}\right)^2 = 456.19.$$

We therefore need at least 457 tires. Now

$$457 \times 25 = 11,425 > 10,000.$$

So, they cannot stay within the budget.

Question 2. (*Signal-to-Noise Ratio: Consistency and MLE*)

The signal-to-noise ratio of a measurement is defined to be μ/σ , where μ and σ are, respectively, the mean and standard deviation of the measurement. Suppose $X_1, \dots, X_n$ form an independent sample from a population with mean μ and variance σ^2 .

a. Find a consistent estimator of signal-to-noise ratio based on $X_1, \dots, X_n$. Carefully justify each step of your proof. **(5 marks)**

Solution:

Let

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \quad S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2,$$

be, respectively, the sample mean and sample variance. Using the law of the large numbers (LLN), $\bar{X}_n \xrightarrow{p} \mu$, i.e. $\bar{X}_n$ is a consistent estimator of μ . We cannot directly apply the LLN to the sample variance. However, we have

$$S_n^2 = \frac{1}{n-1} \sum_{i=1}^n X_i^2 - \frac{n}{n-1} \bar{X}_n^2.$$

Now using the LLN, the first term on the right hand side converges to $\mathbb{E}(X^2)$, while using the LLN and the continuous mapping theorem, the second term converges to $[\mathbb{E}(X)]^2$. Thus $S_n^2 \xrightarrow{p} \sigma^2$. Using the continuous mapping theorem once again, we have $S_n \xrightarrow{p} \sigma$. Note that $n/(n-1) \rightarrow 1$. Given that $\bar{X}_n$ and S_n are, respectively, consistent estimators of μ and σ , using properties of convergence in probability discussed in class their ratio is a consistent estimator of the signal-to-noise ration, i.e.

$$\frac{\bar{X}_n}{S_n} \xrightarrow{p} \frac{\mu}{\sigma}.$$

b. Now suppose that $X_i \stackrel{iid}{\sim} N(\mu, \sigma^2)$. Find the MLE of μ/σ . **(5 marks)**

Solution:

As shown in the lectures, $\bar{X}_n$ and $S_{n,*}^2 = \sum_{i=1}^n (X_i - \bar{X}_n)^2/n$ are respectively the MLE of μ and σ^2 . Using the invariance property of ML method $\bar{X}/S_{n,*}$ is the MLE of μ/σ .

Question 3. (*Signal-to-Noise ratio: MVUE*)

a. Show that

$$\frac{\bar{X}_n}{S_n} = \frac{1}{\sqrt{n}}T + \frac{\mu}{S_n},$$

$$\text{where } T = \frac{\sqrt{n}(\bar{X}_n - \mu)/\sigma}{\sqrt{\frac{U}{(n-1)}}} \sim t_{(n-1)} \quad \text{and} \quad U = \frac{(n-1)S_n^2}{\sigma^2}$$

(5 marks)

Solution:

Adding and subtracting μ we have

$$\frac{\bar{X}_n}{S_n} = \frac{\bar{X}_n - \mu}{S_n} + \frac{\mu}{S_n}$$

Dividing top and bottom of the first term on the right hand side by $\sigma/\sqrt{n}$ and noticing that

$$(n-1)S_n^2 = \sum_{i=1}^n (X_i - \bar{X}_n)^2$$

we obtain the desired result. As shown in the lectures, $T \sim t_{(n-1)}$ if

$$X_i \stackrel{iid}{\sim} N(\mu, \sigma^2) \quad i = 1, 2, \dots, n.$$

b. Suppose $X_i \stackrel{iid}{\sim} N(\mu, \sigma^2)$, $i = 1, 2, \dots, n$. Find $\mathbb{E}\left(\frac{\bar{X}_n}{S_n}\right)$. **(5 marks)**

Solution:

Having noticed that $\mathbb{E}(T) = 0$, we have

$$\mathbb{E}\left(\frac{\bar{X}_n}{S_n}\right) = \mu \mathbb{E}\left(\frac{1}{S_n}\right).$$

Recall that $V = (n-1)S_n^2/\sigma^2 \sim \chi_{n-1}^2$. Thus

$$\begin{aligned} \mathbb{E}(S_n^{-1}) &= \frac{\sqrt{n-1}}{\sigma} \mathbb{E}(V^{-1/2}) \\ &= \frac{\sqrt{n-1}}{\sigma} \int_0^\infty v^{-1/2} \frac{1}{2^{\frac{n-1}{2}} \Gamma(\frac{n-1}{2})} v^{\frac{n-1}{2}-1} e^{-\frac{v}{2}} dv \\ &= \frac{\sqrt{n-1}}{\sigma} \frac{1}{2^{\frac{n-1}{2}} \Gamma(\frac{n-1}{2})} \int_0^\infty v^{\frac{n-2}{2}-1} e^{-\frac{v}{2}} dv \\ &= \frac{\sqrt{n-1}}{\sigma} \frac{1}{2^{\frac{n-1}{2}} \Gamma(\frac{n-1}{2})} 2^{\frac{n-2}{2}} \Gamma(\frac{n-2}{2}) \\ &= \sqrt{\frac{n-1}{2}} \frac{\Gamma(\frac{n-2}{2})}{\Gamma(\frac{n-1}{2})} \frac{1}{\sigma} = \frac{c_n}{\sigma}. \end{aligned}$$

Thus

$$\mathbb{E}\left(\frac{\bar{X}_n}{S_n}\right) = \mu \mathbb{E}\left(\frac{1}{S_n}\right) = c_n \frac{\mu}{\sigma}.$$

c. Use part (b) to find the MVUE of the signal-to-noise ratio. **(5 marks)**

Solution:

It seems natural to start with $\bar{X}_n/S_n$ which is a function of the complete sufficient statistic $(\bar{X}_n, S_n^2)$, and see if we can adjust the estimator to come up with an unbiased estimator if this sample counterpart is biased. Note that using part (a)

$$\frac{\bar{X}_n}{S_n} = \frac{1}{\sqrt{n}}T + \frac{\mu}{S_n},$$

where

$$T = \frac{\sqrt{n}(\bar{X}_n - \mu)/\sigma}{\frac{1}{\sqrt{(n-1)}}\sqrt{(n-1)S_n^2/\sigma^2}} \sim t_{(n-1)},$$

and using part (b)

$$\mathbb{E}\left(\frac{\bar{X}_n}{S_n}\right) = c_n \frac{\mu}{\sigma}.$$

Thus

$$\frac{1}{c_n} \frac{\bar{X}_n}{S_n}$$

where

$$c_n = \sqrt{\frac{n-1}{2}} \frac{\Gamma(\frac{n-2}{2})}{\Gamma(\frac{n-1}{2})}$$

is an unbiased estimator of the signal-to-noise ratio. Given that this estimator is a function of the complete-sufficient statistic $(\bar{X}_n, S_n^2)$, it is the UMVUE.

Question 4. (*Hardy-Weinberg principle*)

The Hardy-Weinberg principle states that allele and genotype frequencies in a population will remain constant from generation to generation in the absence of other evolutionary influences. In the simplest case of a single locus with two alleles, say A and a with probabilities $P(A)=\theta$ and $P(a)=1-\theta$, respectively, the genotype probabilities are $P(AA)=\theta^2$ for the AA homozygotes, $P(aa)=(1-\theta)^2$ for the aa homozygotes, and $P(Aa)=2\theta(1-\theta)$ for the heterozygotes. The genotype proportions θ^2 , $2\theta(1-\theta)$ and $(1-\theta)^2$ are called the Hardy-Weinberg proportions. Let $Y_1, Y_2, \dots, Y_n$ denote a random sample from the following mass function

$$f_\theta(y) = \begin{cases} \theta^2, & \text{if } y = 1, \\ 2\theta(1-\theta), & \text{if } y = 2, \\ (1-\theta)^2 & \text{if } y = 3. \end{cases}$$

Let N_i denote the number of observations equal to i for $i = 1, 2, 3$.

a. Find a non-trivial sufficient statistic for θ . **(7 marks)**

Solution:

Define

$$\gamma(Y) = \begin{cases} 1, & \text{if } Y = 1, \\ 0, & \text{otherwise,} \end{cases} \quad \text{and} \quad \delta(Y) = \begin{cases} 1, & \text{if } Y = 2, \\ 0, & \text{otherwise.} \end{cases}$$

Then $P(Y = y) = [\theta^2]^{\gamma(y)} [2\theta(1-\theta)]^{\delta(y)} [(1-\theta)^2]^{1-\gamma(y)-\delta(y)}$. The likelihood of a random sample $Y_1, Y_2, \dots, Y_n$ is

$$\begin{aligned} \mathcal{L}(\theta; y_1, y_2, \dots, y_n) &= \prod_{i=1}^n P(Y = y_i) \\ &= \prod_{i=1}^n [\theta^2]^{\gamma(y_i)} [2\theta(1-\theta)]^{\delta(y_i)} [(1-\theta)^2]^{1-\gamma(y_i)-\delta(y_i)} \\ &= [\theta^2]^{\sum_{i=1}^n \gamma(y_i)} [2\theta(1-\theta)]^{\sum_{i=1}^n \delta(y_i)} [(1-\theta)^2]^{n - \sum_{i=1}^n \gamma(y_i) - \sum_{i=1}^n \delta(y_i)} \\ &= [\theta^2]^{N_1} [2\theta(1-\theta)]^{N_2} [(1-\theta)^2]^{n - N_1 - N_2} \\ &= 2^{N_2} \theta^{2N_1 + N_2} (1-\theta)^{2n - (2N_1 + N_2)} \\ &= 2^{N_2} (1-\theta)^{2n} \left(\frac{\theta}{1-\theta} \right)^T, \end{aligned}$$

where $T = 2N_1 + N_2$. Using Fisher-Neyman Factorization Theorem, T is a sufficient statistic for θ .

b. Find the MLE of θ .

(8 marks)

Solution:

Taking logarithm of the likelihood we obtain

$$\begin{aligned}\ell(\theta) &= \ell(\theta, y_1, \dots, y_n) = \log \mathcal{L}(\theta, y_1, \dots, y_n) \\ &= N_2 \log 2 + (2N_1 + N_2) \log \theta + [2n - (2N_1 + N_2)] \log(1 - \theta).\end{aligned}$$

Taking derivative with respect to θ , we obtain

$$\frac{d\ell(\theta)}{d\theta} = \frac{2N_1 + N_2}{\theta} - \frac{2n - (2N_1 + N_2)}{1 - \theta}.$$

The MLE of θ is the solution to $d\ell(\theta)/d\theta = 0$. Solving this equation, we find

$$\hat{\theta}_{ML} = \frac{2N_1 + N_2}{2n} = \frac{N_1}{n} + \frac{1}{2} \frac{N_2}{n}.$$

Note that

$$\frac{d^2\ell(\theta)}{d\theta^2} = -\frac{2N_1 + N_2}{\theta^2} - \frac{2n - (2N_1 + N_2)}{(1 - \theta)^2} < 0,$$

and hence $\hat{\theta}_{ML}$ is a maximizer of $\ell(\theta)$.